

Akshat Gupta

Kolkata, West Bengal, India

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EDUCATION

IIT Bhilai

B.Tech. in Data Science and AI; CGPA: 7.86

Chhattisgarh, India

Aug. 2023 – May 2027

Douglas Memorial Higher Secondary School

ISC: 95.5% ICSE: 97%

West Bengal, India

Apr. 2010 – Apr. 2022

RELEVANT COURSEWORK

Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Computer Organization and Architecture, System Design

TECHNICAL SKILLS

Languages: Java, C++, Python, Go, TypeScript, JavaScript

Frontend: Tailwind CSS, React, Next.js

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, PostgreSQL, MySQL

Production & DevOps: Docker, GitHub Actions, Vercel, Render, CI/CD

OPEN SOURCE CONTRIBUTIONS

Apicurio Registry | *Java, REST APIs, JUnit*

- Fixed a **graph traversal** bug that silently dropped nodes from the REST reference-graph endpoint [[merged PR](#)]

Cognee: AI Memory Platform for Agents | *Python, TypeScript, Pytest, CI*

- Fixed **session lock starvation** so a failed run no longer blocks every run queued behind it [[merged PR](#)]
- Built the version and **npm update-hint** surface for the OpenClaw plugin, now shipping on npm [[PR](#)]

PROJECTS

AI-Powered Learning Assistant

[GitHub](#) | [Live](#)

- Kept AI answers grounded in source content by building a **MERN** platform that retrieves only the relevant parts of an uploaded PDF for chat, summaries, flashcards, and quizzes
- Selected the production review scheduler by implementing **SM-2** and **FSRS** spaced-repetition algorithms and simulating how efficiently each one schedules reviews
- Secured multi-user accounts with **JWT auth** using refresh token rotation, account lockout, and a bring-your-own-key option so users run AI features on their own OpenAI key
- Caught regressions before release with **CI/CD** running automated tests, dependency vulnerability scans, and Playwright end-to-end tests against a live backend

Pond Catchment Analysis

[GitHub](#)

- Translated hydrology requirements into a **FastAPI geospatial pipeline** processing KML/KMZ contours into terrain grids and pond catchments via surface-flow routing
- Automated **pond-site selection** by modeling drainage networks, slope, catchment size, and elevation constraints
- Exposed runoff and stage-storage GeoJSON layers over REST endpoints, integrating **10 years of rainfall data** for downstream mapping clients
- Verified pipeline correctness with **451 automated tests**, including mass-balance and analytical checks

Multi-Agent Research Assistant

[GitHub](#)

- Guaranteed termination on ambiguous queries by coordinating 4 **LangGraph** agents on a shared state graph with a 3-pass cap
- Cut redundant LLM calls by adding an **intent classifier** that routes follow-ups to direct answers or in-place edits
- Gave step-level **observability** into long-running jobs by streaming agent progress over **FastAPI SSE**
- Shipped a **Next.js** UI with PDF/CSV ingestion, web search, and PDF export

ACHIEVEMENTS

- Selected as a **Reliance Foundation Scholar** through a national merit-based scholarship program
- Selected for **FOSS Overflow**, an open source mentorship program run by OpenLake at IIT Bhilai
- Solved **1000+** problems in Java across competitive programming platforms including [Codeforces](#) and [LeetCode](#)